

JOB MARKET ANALYSIS

An End-to-End SQL & Power BI Project on the U.S. Data & Analytics Job Market

Prepared by:

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Data Source:

Glassdoor Jobs Dataset

Database:

Job_Market_Analysis

Tools Used:

MySQL Workbench (SQL), Power BI Desktop (DAX & Dashboards)

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Table of Contents

- 1. Executive Summary..... 3
- 2. Introduction 4
- 3. Objectives 4
- 4. Dataset Description..... 5
- 5. Technology Stack 6
- 6. Methodology 7
 - 6.1 Data Cleaning Process 7
 - 6.2 SQL Analysis 7
 - 6.3 Power BI Data Modeling..... 7
- 7. Executive Dashboard 9
- 8. Salary Dashboard 11
- 9. Skills Dashboard..... 13
- 10. Company Dashboard..... 15
- 11. Key Business Insights 17
- 12. Business Recommendations..... 18
- 13. Challenges Faced 19
- 14. Project Outcomes 20
- 15. Skills Demonstrated 20
- 16. Future Enhancements 21
- 17. References 21
- 18. Appendix..... 22
- 19. Conclusion 30

1. Executive Summary

This report documents an end-to-end data analytics project covering the U.S. data and analytics job market, built on a Glassdoor job-postings dataset. The project combines SQL-based data cleaning and exploratory analysis (MySQL Workbench) with a four-page interactive Power BI dashboard suite — Executive, Salary, Skills, and Company — to translate raw job-posting records into decision-ready business insight.

The analysis addresses four central questions: where is demand for data roles concentrated, how much do these roles pay and what drives pay variation, which technical skills are most requested, and what kind of organizations are hiring. The resulting dashboards cover 2,000 job listings across 318 companies, 35 states, and 57 industries, with an average posted salary(K) of \$107.

Key takeaways include a market dominated by the Data Scientist title, a concentration of openings in Virginia, Texas, and Pennsylvania, a technical baseline of Python, Excel, and SQL, and a measurable pay premium at publicly traded and mid-sized companies. These findings, along with full methodology, dashboards, and supporting SQL code, are presented in the sections that follow.

2,000 Job Listings	318 Companies	35 States	\$107 Avg. Salary(K)
220 Unique Job Titles	57 Industries	392 Python Job Postings	3.77 Avg. Company Rating

2. Introduction

The data and analytics job market has grown rapidly over the past decade, with organizations across nearly every industry now hiring for roles such as Data Scientist, Data Engineer, and Data Analyst. For job seekers, recruiters, and workforce planners, understanding this market — where the jobs are, what they pay, what skills they demand, and who is hiring — is essential to making informed decisions.

This project analyzes a real-world Glassdoor job-postings dataset to answer exactly those questions. It follows a typical analytics workflow: raw data is imported into a relational database (MySQL), cleaned and validated with SQL, and then modeled and visualized in Power BI to produce an interactive, business-ready dashboard suite. The result is a practical decision-support tool that also demonstrates a complete SQL-to-Power BI analytics workflow.

3. Objectives

This project was undertaken with the following objectives:

- **1.** Clean and validate the raw Glassdoor job-postings dataset, removing duplicates and standardizing placeholder/missing values.
- **2.** Use SQL to explore ten core business questions covering geography, compensation, industry, company, and skills.
- **3.** Design a Power BI data model with derived fields (state, seniority, skill flags) to support flexible, cross-filterable analysis.
- **4.** Translate analytical findings into actionable business recommendations for job seekers, employers, and workforce planners.
- **5.** Translate the visual and query-based findings into concrete, actionable business recommendations for job seekers, employers, and workforce planners.

4. Dataset Description

The project is built on a Glassdoor job-postings dataset containing 41 analytical attributes per job posting after data preprocessing. The table below summarizes the primary categories of attributes included in the dataset.

Category	Key Attributes
Job Details	Job Title, Salary Estimate, Job Title Category, Seniority Level, Degree
Location	Location, Headquarters, Job Location
Company Profile	Company Name, Rating, Size, Founded, Age, Ownership Type, Industry, Sector, Revenue, Competitors
Compensation	Lower Salary, Upper Salary, Average Salary(K), Hourly, Employer Provided
Skill Flags (binary)	Python, Spark, AWS, Excel, SQL, SAS, Keras, PyTorch, Scikit-learn, TensorFlow, Hadoop, Tableau, Power BI, Flink, MongoDB, Google Analytics

4.1 Data Quality Notes

- **Placeholder Values.** Uses -1 as a placeholder for unavailable values in Rating, Founded, Industry, Sector, and Competitors — addressed during SQL data cleaning (Section 6.1).
- **Composite Fields.** The Location field combines city and state (e.g., “Albuquerque, NM”) and required string parsing to isolate state-level geography for analysis.
- **Skill Encoding.** Technical skill requirements are stored as individual binary columns (1 = required, 0 = not required) rather than a single delimited list, enabling direct SQL aggregation and Power BI unpivoting.
- **Feature Selection.** The original dataset contained 42 columns. The non-essential FIELD column was removed during preprocessing, resulting in a final dataset of 41 analytical attributes.

5. Technology Stack

The project uses a standard, industry-representative analytics stack, moving data from raw file to relational database to interactive dashboard:

Tool / Technology	Role in the Project
MySQL Workbench	Relational database for storing the cleaned dataset; used for data cleaning, validation, and exploratory SQL analysis.
SQL (DDL / DML / Aggregate Functions)	Database creation, duplicate detection, NULL-value validation, placeholder cleanup, and business-question analysis using SQL queries.
Power BI Desktop	Data modeling, interactive dashboard development, DAX measure creation, and report visualization.
Power Query (M)	ETL layer inside Power BI — shaping, transforming, and unpivoting fields (e.g., skill flags) ahead of visualization.
DAX (Data Analysis Expressions)	Created calculated measures and KPIs, including average salary, median salary, job counts, and percentage-based metrics used throughout the dashboards.
CSV (Glassdoor Export)	Primary raw data source used throughout the SQL and Power BI analytics pipeline.

6. Methodology

The project followed a structured three-stage analytics methodology to ensure data quality, analytical accuracy, and reliable dashboard reporting. (1) clean and validate the raw data directly in SQL, (2) run exploratory SQL queries to answer core business questions, and (3) load the cleaned dataset into Power BI and build an analytical data model for interactive dashboard development. Each stage is detailed below.

6.1 Data Cleaning Process

The raw dataset was imported into MySQL Workbench, and a working copy (**Job_Market_Analysis_Clean**) was created to preserve the integrity of the original dataset. Data cleaning followed a structured four-step workflow:

- **1. Duplicate Check.** Row counts and a duplicate check on the ID field confirmed the dataset contained no duplicate job postings.
- **2. Null Audit.** Core fields (Job Title, Company Name, Location, Headquarters, Industry, Sector) were checked for missing values using conditional SUM aggregation.
- **3. Placeholder Cleaning.** The original Glassdoor dataset uses -1 as a placeholder value for unavailable data in fields such as Rating, Industry, Sector, Founded, and Competitors. These were identified and replaced with NULL via targeted UPDATE statements so they would not distort downstream averages and aggregations.
- **4. Validation.** Post-update queries confirmed zero remaining -1 placeholder values across all affected columns before proceeding to analysis.

6.2 SQL Analysis

Following data cleaning, nine business-oriented SQL queries were executed against Job_Market_Analysis_Clean to answer the project's core analytical questions prior to Power BI dashboard development. Each query maps directly to one of the project's guiding business questions.

#	Business Question	Primary SQL Concepts
1	Which states have the highest number of job openings?	SUBSTRING_INDEX + GROUP BY
2	What are the average minimum and maximum salaries by state?	AVG, ROUND, GROUP BY

#	Business Question	Primary SQL Concepts
3	What is the average salary by job location?	AVG, GROUP BY, ORDER BY
4	Which industries post the most data-related jobs?	COUNT, WHERE, GROUP BY
5	Which companies have the most job openings?	COUNT, GROUP BY, LIMIT
6	Which job titles appear most frequently?	COUNT, GROUP BY, LIMIT
7	What is the average salary for the most common job titles?	AVG + COUNT, GROUP BY
8	Which skills are required most often, by job title?	SUM (binary flags), GROUP BY
9	How does average salary relate to required education?	AVG, COUNT, GROUP BY

Full query text for each of the above is reproduced in Appendix A.

The SQL analysis transformed raw job-posting data into structured business insights that were subsequently visualized through interactive Power BI dashboards, ensuring consistency between database analysis and dashboard reporting.

6.3 Power BI Data Modeling

The cleaned dataset was imported into Power BI Desktop, Power Query (M) was used to transform and prepare the dataset for interactive dashboard development:

- **Derived Columns.** Location was split into city and state components, and Founded was converted into a calculated Company Age field, mirroring the SQL-side transformations.
- **Skill Unpivoting.** The sixteen binary skill columns (Python, SQL, Excel, AWS, Spark, Tableau, and others) were unpivoted into a single **Skill / Has_Skill** column pair, enabling the Skills dashboard to filter and rank by any individual skill.
- **DAX Measures.** DAX measures were created for Total Jobs, Average / Median / Highest / Lowest Salary, Average Company Rating, and percentage-of-total calculations feeding the donut and 100%-stacked visuals.
- **Interactivity.** Slicers for Industry, State, Company Name, Job Title, Seniority Level, and Company Size were placed on every page and synced where relevant, allowing users to filter and analyze all four dashboards through synchronized slicers.
- **Report Design.** A consistent navigation bar (Executive / Salary / Skills / Company) with a distinct accent color per page was used to create a consistent and professional user experience across all dashboard pages.

7. Executive Dashboard

The Executive dashboard provides a high-level view of the entire job market: 2,000 postings across 220 unique job titles, 318 companies, 35 states, and 57 industries, with an average salary(K) of \$107.

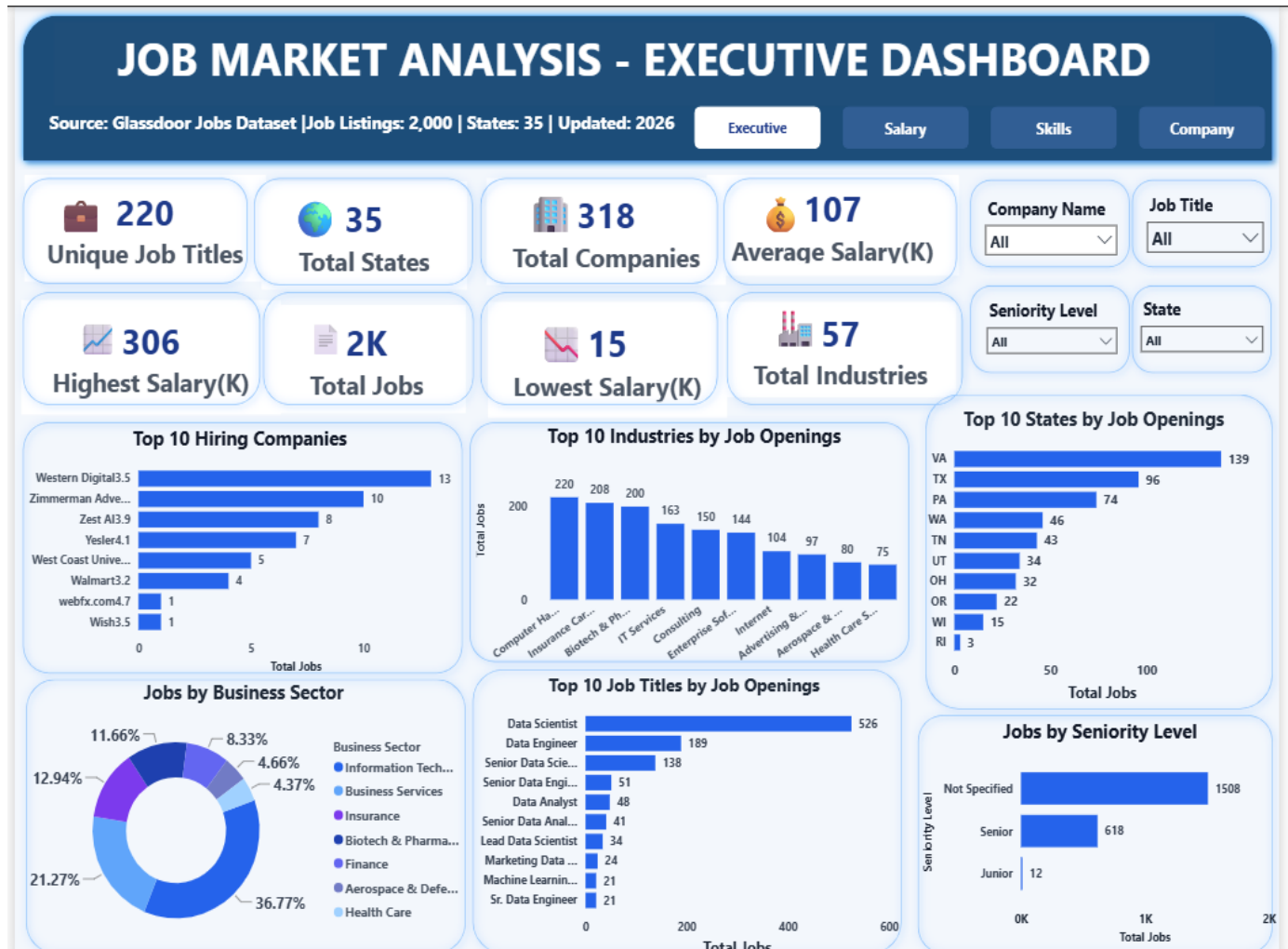


Figure 7.1: Executive Dashboard developed in Power BI, featuring interactive filtering by Industry, State, Company Name, and Job Title.

Key Insights

- **Role concentration:** Data Scientist roles dominate the job market with 526 postings, more than double the number of Data Engineer positions (189). Senior Data Scientist roles (138) complete the top three.
- **Geographic concentration:** Virginia leads all states with 139 job openings, followed by Texas (96) and Pennsylvania (74) — together these three states account for a significant share of overall job demand.

- **Industry concentration:** Computer Hardware (220), Insurance Carriers (208) and Biotech & Pharmaceuticals (200) are the top hiring industries; Information Technology is the largest business sector overall at 36.8% of postings.
- **Seniority distribution:** The majority of job postings are classified under the **Not Specified** seniority category, indicating that many employers do not clearly define experience requirements in their job listings, while Senior and Junior roles account for a smaller share of the market.
- **Top employer:** Western Digital leads individual hiring activity with 13 open roles, followed by Zimmerman Advertising (10) and Zest AI (8).

Table 7.1: Top 5 States by Job Openings

State	Job Openings
Virginia	139
Texas	96
Pennsylvania	74
Washington	46
Tennessee	43

Virginia recorded the highest number of job openings (139), followed by Texas (96) and Pennsylvania (74), indicating stronger hiring demand in these states.

8. Salary Dashboard

The Salary Dashboard provides a comprehensive analysis of compensation trends across 961 high-paying roles, with a median salary of \$104.5K, an average minimum salary of \$80K, and an average maximum salary of \$135K. The resulting salary range of \$55K highlights significant variation across industries, companies, and locations.

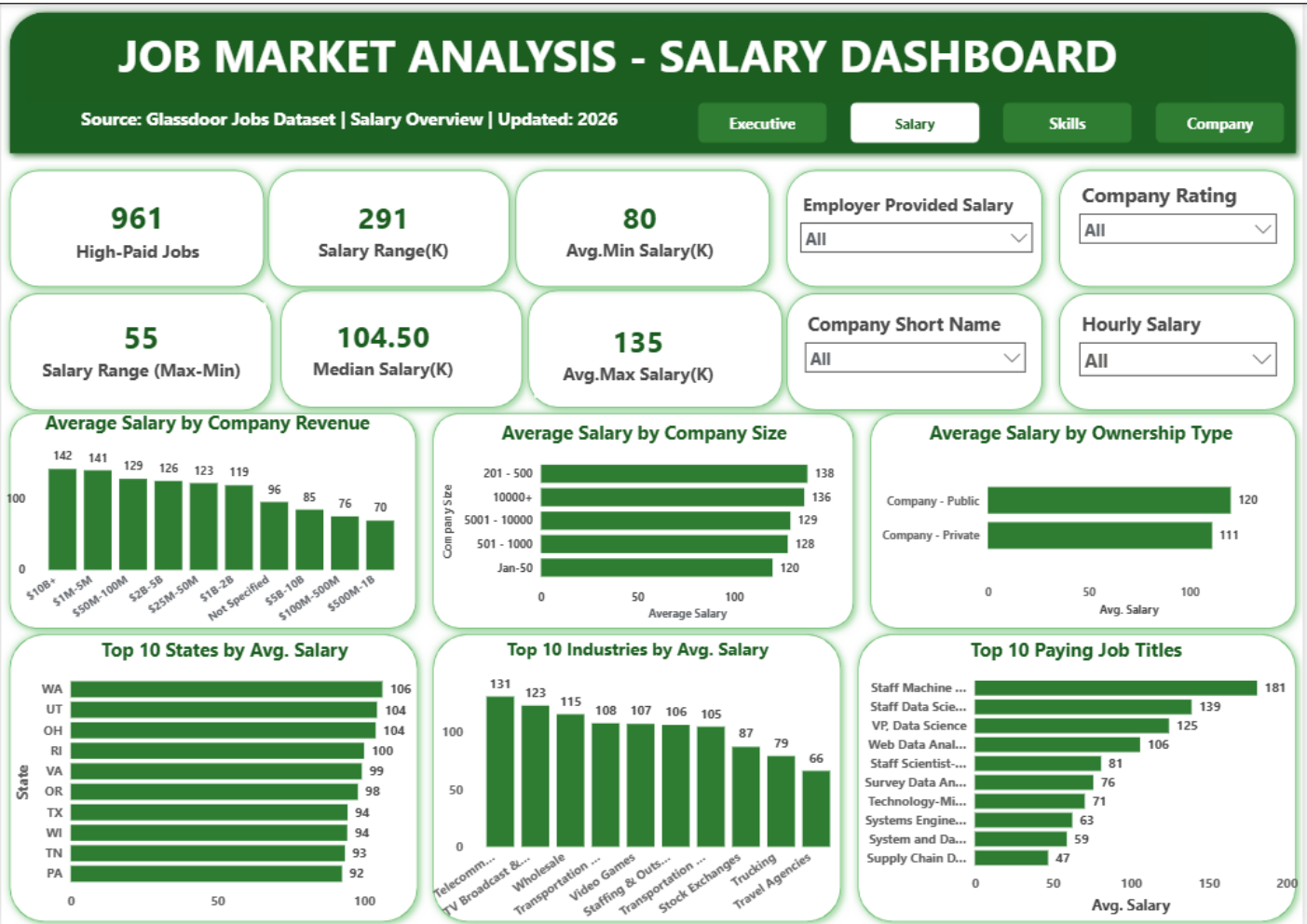


Figure 8.1: Salary Dashboard developed in Power BI, featuring interactive filtering by Industry, State, Company, and Job Title.

Key Insights

- Top-paying states:** Washington (\$106K), Utah (\$104K), and Ohio (\$104K) offer the highest average salaries, outperforming several traditionally high cost-of-living states such as California, suggesting strong demand-driven salary premiums outside the coastal markets.

- **Top-paying industries:** Telecommunications (\$131K), TV Broadcasting & Cable (\$123K) and Wholesale (\$115K) offer the highest average compensation, while Travel Agencies (\$66K) and Trucking (\$79K) sit at the lower end.
- **Company size matters:** Mid-sized companies (201–500 employees) offer the highest average salary (\$138K), narrowly ahead of the largest enterprises (10,000+, \$136K) — small companies (1–50 employees) pay the least (\$120K).
- **Ownership premium:** Publicly traded companies pay an average of \$120K versus \$111K at private companies, a roughly an 8% salary premium for publicly traded companies.
- **Highest-paying titles:** Staff Machine Learning roles top the pay scale (\$181K avg.), followed by Staff Data Scientist (\$139K) and VP of Data Science (\$125K).
- **Salary variability:** The average salary ranges from \$80K (minimum) to \$135K (maximum), resulting in a \$55K average salary spread, indicating substantial variation in compensation across industries, companies, and job roles.

Table 8.1: Top 5 Highest-Paying Job Titles

Job Title	Average Salary (USD)
Staff Machine Learning Engineer	\$181K
Staff Data Scientist	\$139K
VP, Data Science	\$125K
Web Data Analyst	\$106K
Staff Scientist	\$81K

Staff Machine Learning Engineer records the highest average salary (\$181K), followed by Staff Data Scientist (\$139K) and VP, Data Science (\$125K), highlighting the premium placed on advanced machine learning and leadership roles.

9. Skills Dashboard

The Skills Dashboard analyzes demand for technical skills across all job postings. Python (392), Excel (388), and SQL (380) emerge as the most frequently requested skills, followed by AWS (176) and Spark (167), establishing a clear baseline technical skill set for data-related roles.

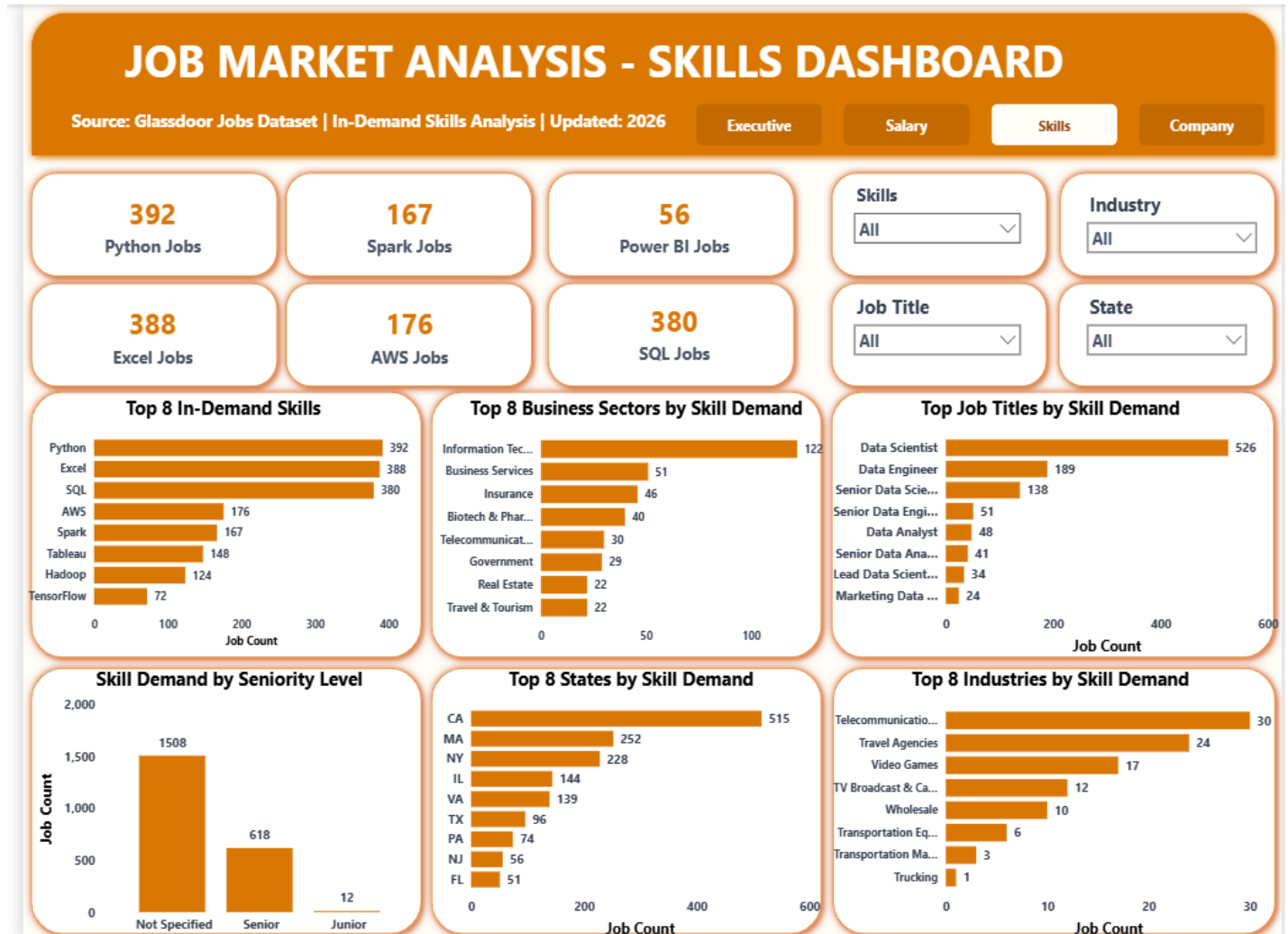


Figure 9.1: Skills Dashboard developed in Power BI, featuring interactive filtering by Industry, State, Company, and Job Title.

Key Insights

- **Core skill trio:** Python, Excel, and SQL are near-universal requirements, each appearing in roughly 19–20% of all postings, confirming they remain the foundational toolkit for data professionals.

- **Cloud & big data:** AWS (176) and Spark (167) show strong demand for cloud and distributed-computing skills, while Tableau (148) and Hadoop (124) remain relevant for visualization and big-data engineering.
- **Emerging skills:** TensorFlow (72) and Power BI (56) show comparatively lower but meaningful demand, highlighting specialized requirements in machine learning and business intelligence roles.
- **Role demand:** Data Scientist (526 postings), Data Engineer (189), and Senior Data Scientist (138) account for the highest demand for technical skills, highlighting their continued dominance within the data analytics and AI job market.
- **Geographic skill hubs:** California (515), Massachusetts (252) and New York (228) are the leading states for skilled data roles, together accounting for a large share of national skill demand.
- **Sector demand:** Information Technology (122 postings) is by far the leading sector for skill-tagged roles, more than double Business Services (51), the second-highest sector.

Table 9.1: Top 5 In-Demand Skills

Skill	Job Postings
Python	392
Excel	388
SQL	380
AWS	176
Spark	167

Python, Excel, and SQL are the three most frequently requested technical skills, appearing in nearly one-fifth of all job postings, highlighting their importance as core competencies for data professionals.

10. Company Dashboard

The Company Dashboard provides an overview of 318 hiring organizations, including company ratings, organizational age, ownership structure, revenue categories, and company size. Among these organizations, 107 are highly rated, with an average company age of 155 years and an average Glassdoor rating of 3.77 out of 5.

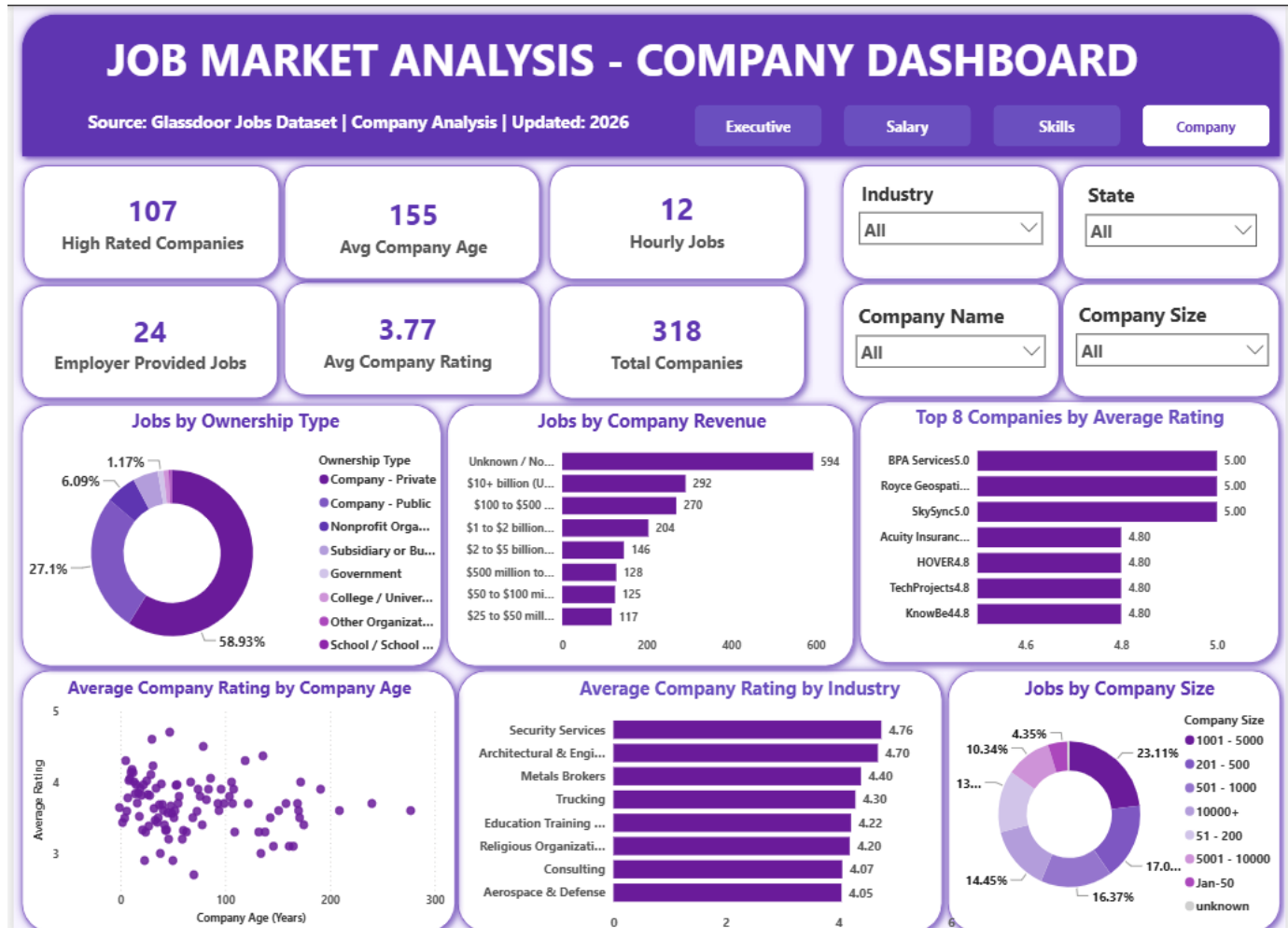


Figure 10.1: Company Dashboard developed in Power BI, featuring interactive filtering by Industry, State, Company Name, and Company Size.

Key Insights

- **Ownership structure.** Privately held companies account for 58.9% of job postings, more than double publicly traded companies (27.1%), with nonprofits (6.1%) and other structures making up the remainder.
- **Revenue transparency gap.** 594 postings come from companies with unknown or undisclosed revenue — the single largest category — followed by large enterprises earning \$10B+ (292 postings).

- **Top-rated employers.** Top-rated employers. BPA Services, Royce Geospatial, and SkySync each have an average company rating of 5.0, while Acuity Insurance, HOVER, TechProjects, and KnowBe4 follow closely with ratings of 4.8.
- **Company size distribution.** Mid-to-large companies (1,001–5,000 employees) post the most jobs at 23.1% of the total, ahead of smaller firms (201–500, 17.0%) and large enterprises (501–1,000, 16.4%).
- **Rating by industry.** Security Services (4.76) and Architectural & Engineering firms (4.70) receive the highest average employee ratings among industries with sufficient review volume.
- **Company maturity.** Company ratings remain relatively consistent across different company ages, suggesting that organizational age alone is not a strong indicator of employee satisfaction.

Table 10.1: Top 5 Companies by Average Rating

Company	Average. Rating
BPA Services	5.0
Royce Geospatial	5.0
SkySync	5.0
Acuity Insurance	4.80
HOVER	4.80

BPA Services, Royce Geospatial, and SkySync achieved the highest average company rating (5.00), while Acuity Insurance and HOVER followed closely with ratings of 4.80, indicating consistently high employee satisfaction among these organizations.

11. Key Business Insights

Bringing together the SQL findings and all four dashboards, six cross-cutting insights define the current state of the data and analytics job market:

Dimension	Key Finding
Roles	Data Scientist is the dominant title (526 postings, ~25% of the market).
Geography	Virginia, Texas, and Pennsylvania are the top three states for job volume; California, Massachusetts and New York lead in skill demand.
Compensation	Median salary is \$104.5K; Telecommunications and top-paying states such as Washington and Utah offer the highest average pay.
Skills	Python, Excel, and SQL are the most frequently requested technical skills, appearing in approximately one in five job postings.
Company Profile	Privately held companies account for the majority of job postings, while publicly traded companies offer approximately 8% higher average salaries.
Seniority	Most job postings do not specify a seniority level, while only a small number explicitly target junior candidates.

12. Business Recommendations

For Job Seekers

- Prioritize proficiency in Python, SQL, and Excel, as these appear in the largest share of postings regardless of seniority or industry.
- Target Virginia, Texas, and Pennsylvania for the highest concentration of open roles, or California, Massachusetts, and New York for the strongest overall skill demand.
- Consider opportunities in Telecommunications, TV Broadcasting & Cable, and Wholesale industries, as well as publicly traded and mid-sized (201–500 employees) companies, which offer above-average compensation.

For Employers

- Clarify seniority levels in job postings, as many listings do not explicitly specify experience requirements. This can improve applicant matching and attract suitable entry-level candidates.
- Benchmark salaries against public-company and mid-size-company averages (\$136K–\$138K), which currently outpace both very large and very small companies.
- Disclose revenue and company-size data where possible — nearly a third of postings currently fall into an “unknown revenue” bucket, limiting candidates' ability to evaluate employer stability.

For Workforce Planners

- Focus workforce development and training programs on cloud (AWS) and distributed computing (Spark) skills, which show strong demand but lower saturation than Python/SQL.
- Encourage more structured junior and entry-level hiring initiatives to strengthen the future talent pipeline, as relatively few job postings explicitly target junior candidates.

These recommendations are derived from the SQL analysis and Power BI dashboards and are intended to support informed decision-making for job seekers, employers, and workforce planners.

13. Challenges Faced

- **Placeholder Values Distorting Metrics.** The Glassdoor dataset used -1 as a placeholder for missing Rating, Founded, Industry, Sector, and Competitors values; left unaddressed, these would have silently skewed every average and ranking. Resolved with targeted SQL UPDATE statements converting placeholders to NULL.
- **Composite Location Field.** The Location field combined city and state in a single text string (e.g., “Albuquerque, NM”), requiring SUBSTRING_INDEX parsing in SQL and an equivalent split transformation in Power Query to isolate state-level geography.
- **Skill Data Reshaping.** Required skills were encoded as sixteen separate binary columns rather than a single list, which was efficient for SQL SUM aggregation but needed an unpivot transformation in Power Query before it could drive a single, filterable Skills dashboard.
- **Incomplete Categorical Data.** Many job postings did not specify a seniority level, and a large share of company revenue information was marked as unknown. These limitations reduced the level of detail available for certain analyses and were documented as findings rather than corrected.
- **Balancing Dashboard Scope.** Presenting four distinct analytical perspectives (Executive, Salary, Skills, Company) without duplicating content or overwhelming the reader required careful curation of which metrics belonged on which page.
- **Data Validation Across SQL and Power BI.** SQL query outputs were cross-verified with Power BI measures and dashboard visualizations to ensure consistency, accuracy, and reliability throughout the analytical workflow.

Despite these challenges, the project successfully produced a clean analytical dataset and interactive Power BI dashboards, enabling reliable analysis, meaningful business insights, and actionable recommendations.

14. Project Outcomes

The project delivered a complete, reusable analytics package:

- **Clean Dataset.** A validated MySQL database (Job_Market_Analysis_Clean) with all duplicate, null, and placeholder issues resolved and documented.
- **SQL Query Library.** Ten reusable SQL queries designed to answer the project's core business questions, with complete query scripts provided in Appendix A.
- **Interactive Dashboard.** A four-page, cross-filterable Power BI dashboard suite (Executive, Salary, Skills, Company) covering 2,000 job listings.
- **Project Documentation.** Comprehensive project documentation, including methodology, dashboard explanations, business insights, recommendations, and appendices, to support future maintenance and knowledge transfer.
- **Business Report.** This document consolidates the project methodology, visualizations, and business insights into a single shareable deliverable for stakeholders who may not have direct access to Power BI.

15. Skills Demonstrated

Technical Skills

- **SQL.** Database design, DDL/DML, aggregate functions, string functions, conditional aggregation, and iterative data validation.
- **Data Cleaning.** Systematic detection and correction of duplicates, nulls, and placeholder values.
- **Data Validation.** Cross-verifying SQL query outputs with Power BI measures and dashboard visualizations to ensure analytical accuracy and consistency.
- **Power BI.** Power Query (M) transformations, data modeling, and DAX measure design.
- **Dashboard Design.** Multi-page dashboard layout, slicer design, color theory, and consistent visual branding.

Analytical & Communication Skills

- **Business Analysis.** Translating open-ended business questions into structured, testable queries and visuals.
- **Data Storytelling.** Identifying patterns across multiple dashboards and synthesizing them into a coherent narrative.
- **Technical Documentation.** Producing a structured, professional report suitable for non-technical stakeholders.

16. Future Enhancements

- **Predictive Salary Modeling.** Apply regression or machine-learning models to predict expected salary ranges based on job title, location, company size, and required skills.
- **NLP-Based Skill Extraction.** Use natural-language processing on raw job descriptions to extract a richer, more granular skill taxonomy beyond the current sixteen flagged skills.
- **Real-Time Data Refresh.** Connect the Power BI model to a live job-postings API or scheduled scrape to keep the dashboard continuously updated instead of relying on static data exports.
- **Geographic Map Visualization.** Add a choropleth map visual to make state-level job and salary concentration immediately visible at a glance.
- **Cross-Platform Data Integration.** Incorporate postings from additional platforms (e.g., LinkedIn, Indeed) to validate findings and reduce single-source bias.
- **Advanced SQL Analytics & Trend Analysis.** Implement SQL window functions, ranking, and trend analysis on continuously updated datasets to provide deeper hiring and salary insights.

17. References

- **Glassdoor Jobs Dataset (Kaggle).** Primary dataset used for the Job Market Analysis project.
- **MySQL 8.0 Reference Manual.** Official documentation for SQL syntax, data types, and query optimization.
- **Microsoft Power BI Documentation.** Official documentation for Power BI Desktop, Power Query, data modeling, and report design.
- **Microsoft Power Query Documentation.** Official documentation for data transformation, data cleaning, and data preparation in Power BI.
- **Microsoft DAX Reference.** Official reference guide for DAX functions and calculated measures used in the data model.

18. Appendix

Appendix A: Full SQL Script

The complete SQL script used for database creation, data cleaning, data validation, and exploratory analysis (Job_Market_Analysis_SQL_Queries.sql) is provided below for reference and reproducibility—

Project Name: Job Market Analysis Using SQL

Author : Baliyi Mohan

Database : Job_Market_Analysis

Tool : MySQL Workbench

----- Query 1: Database Creation & Data Import -----

```
CREATE DATABASE Job_Market_Analysis;
USE Job_Market_Analysis;
SELECT * FROM Job_Market_Analysis;
```

NOTE:

```
-- Import the CSV dataset into the Job_Market_Analysis table
-- using MySQL Workbench -> Table Data Import Wizard.
```

----- CREATED DUPLICATE TABLE -----

```
CREATE TABLE Job_Market_Analysis_Clean AS
SELECT *
FROM Job_Market_Analysis;
```

---- Total rows -----

```
SELECT COUNT(*) AS Total_Rows
FROM Job_Market_Analysis_Clean;
```

```
DESCRIBE Job_Market_Analysis_Clean;
```

----- Checking Duplicate -----

```
SELECT
    ID,
    COUNT(*) AS Duplicate_Count
FROM Job_Market_Analysis_Clean;
GROUP BY ID
HAVING COUNT(*) > 1;
```

----- Checking NULL values -----

```
SELECT
    SUM(Job_Title IS NULL) AS Job_Title_Null,
    SUM(Company_Name IS NULL) AS Company_Name_Null,
    SUM(Location IS NULL) AS Location_Null,
    SUM(Headquarters IS NULL) AS Headquarters_Null,
    SUM(Industry IS NULL) AS Industry_Null,
    SUM(Sector IS NULL) AS Sector_Null
FROM Job_Market_Analysis_Clean;
```

----- Check -1 Values -----

```
SELECT
    SUM(Founded = -1) AS Founded_Minus1,
    SUM(Rating = -1) AS Rating_Minus1,
    SUM(Industry = '-1') AS Industry_Minus1,
    SUM(Sector = '-1') AS Sector_Minus1,
    SUM(Competitors = '-1') AS Competitors_Minus1
FROM Job_Market_Analysis_Clean;
```

----- Data Cleaning -----

```
DESCRIBE Job_Market_Analysis_Clean;
```

-- Replace -1 in Rating, Industry, Sector, Founded, Competitors -

```
UPDATE Job_Market_Analysis_Clean
SET Rating = NULL
```

```

WHERE Rating = -1;
UPDATE Job_Market_Analysis_Clean
SET Industry = NULL
WHERE Industry = '-1';
UPDATE Job_Market_Analysis_Clean
SET Sector = NULL
WHERE Sector = '-1';
UPDATE Job_Market_Analysis_Clean
SET Founded = NULL
WHERE Founded = -1;
UPDATE Job_Market_Analysis_Clean
SET Competitors = NULL
WHERE Competitors = '-1';

```

----- Verify the Updates -----

```

SELECT
    SUM(Founded = -1) AS Founded_Minus1,
    SUM(Rating = -1) AS Rating_Minus1,
    SUM(Industry = '-1') AS Industry_Minus1,
    SUM(Sector = '-1') AS Sector_Minus1,
    SUM(Competitors = '-1') AS Competitors_Minus1
FROM Job_Market_Analysis_Clean;

```

Exploratory Data Analysis (EDA)

Business Analysis Queries

---- Query 2: States with Most Number of Jobs ---

```

SELECT
    SUBSTRING_INDEX(Location, ',', -1) AS State,
    COUNT(*) AS Total_Jobs
FROM Job_Market_Analysis_Clean
GROUP BY State
ORDER BY Total_Jobs DESC;

```

---- Top 10 States ---

```
SELECT
    SUBSTRING_INDEX(Location, ',', -1) AS State,
    COUNT(*) AS Total_Jobs
FROM Job_Market_Analysis_Clean
GROUP BY State
ORDER BY Total_Jobs DESC
LIMIT 10;
```

---- Query 3: Average Minimal and Maximal Salaries in Different States.

```
SELECT
    SUBSTRING_INDEX(Location, ',', -1) AS State,
    ROUND(AVG(Lower_Salary),2) AS Avg_Min_Salary,
    ROUND(AVG(Upper_Salary),2) AS Avg_Max_Salary
FROM Job_Market_Analysis_Clean
GROUP BY State
ORDER BY Avg_Max_Salary DESC;
```

---- Query 4: Average Salary in Different States.

```
SELECT
    Job_Location,
    ROUND(AVG(Avg_SalaryK),2) AS Average_Salary_K
FROM Job_Market_Analysis_Clean
GROUP BY Job_Location
ORDER BY Average_Salary_K DESC;
```

--- Query 5: Top 5 Industries with Maximum Number of Data Science Related Job Postings.

```
SELECT
    Industry,
    COUNT(*) AS Total_Job_Postings
FROM Job_Market_Analysis_Clean
WHERE Industry IS NOT NULL
    AND Industry <> ''
GROUP BY Industry
ORDER BY Total_Job_Postings DESC
LIMIT 5;
```

---- Query 6: Companies with Maximum Number of Job Openings.

```
SELECT
    Company_Name,
    COUNT(*) AS Total_Job_Postings
FROM Job_Market_Analysis_Clean
GROUP BY Company_Name
ORDER BY Total_Job_Postings DESC
LIMIT 10;
```

--- Query 7: Job Titles with Most Number of Jobs.

```
SELECT
    Job_Title,
    COUNT(*) AS Total_Job_Postings
FROM Job_Market_Analysis_Clean
GROUP BY Job_Title
ORDER BY Total_Job_Postings DESC
LIMIT 10;
```

---- Query 8: Salary of Job Titles with Most Number of Jobs.

```
SELECT
    Job_Title,
    ROUND(AVG(Avg_SalaryK), 2) AS Average_Salary_K,
    COUNT(*) AS Total_Job_Postings
FROM Job_Market_Analysis_Clean
GROUP BY Job_Title
ORDER BY Total_Job_Postings DESC;
```

----- Query 9: Skills Required by Companies for Each Job Title.

```
SELECT
    Job_Title,
    SUM(Python) AS Python_Jobs,
    SUM(SQL_) AS SQL_Jobs,
    SUM(Excel) AS Excel_Jobs,
    SUM(Tableau) AS Tableau_Jobs
FROM Job_Market_Analysis_Clean
GROUP BY Job_Title
ORDER BY COUNT(*) DESC;
```

---- Query 10: Relation between Average Salary and Education.

```
SELECT
    Degree,
    ROUND(AVG(Avg_SalaryK), 2) AS Average_Salary_K,
    COUNT(*) AS Total_Job_Postings
FROM Job_Market_Analysis_Clean
GROUP BY Degree
ORDER BY Average_Salary_K DESC;
```

End of Job Market Analysis SQL Project

Appendix B: Dataset Column Dictionary

Column	Description
ID	Unique identifier for each job posting.
Job_Title / job_title_sim	Raw and standardized job title.
Salary_Estimate	Original Glassdoor-provided salary range text.
Rating	Company rating out of 5 (placeholder -1 cleaned to NULL).
Company_Name / company_txt	Employer name (original and cleaned company name).
Location / Job_Location	Job posting location (city/state) and derived state information.
Headquarters	City/state of company headquarters.
Size	Employee headcount band (e.g., 501–1000).
Founded / Age	Year founded and derived company age in years.
Type_of_ownership	Ownership structure (Private, Public, Nonprofit, etc.).
Industry / Sector	Industry and broader business sector classification.
Revenue	Annual revenue band, where disclosed.
Competitors	Named competitor companies, where disclosed.
Lower_Salary / Upper_Salary / Avg_SalaryK	Minimum, maximum, and average estimated salary (in thousands).
Hourly / Employer_provided	Flags for hourly-wage and employer-provided salary postings.

Column	Description
Binary skill indicator columns (e.g., Python, SQL, Excel, AWS, Spark, Tableau, Hadoop, MongoDB, Power BI etc.)	Binary flags (1/0) indicating whether each skill is required.
seniority_by_title	Seniority level inferred from job title (Junior / Senior / Not Specified).
Degree	Minimum education requirement (e.g., Bachelor's, Master's).

19. Conclusion:

This project demonstrates a complete, real-world analytics workflow — from a raw CSV dataset to a cleaned relational database to a polished, interactive Power BI dashboard suite.

The analysis shows that the modern data-science job market, while broad in title and location, is concentrated around a core set of roles, states, and skills.

The analysis indicates that Data Scientist remains the flagship job title, while SQL and Excel continue to be among the most frequently requested technical skills, and compensation appears to be influenced more by ownership structure and industry than by company size alone.

Beyond the specific findings, the project illustrates the value of pairing SQL's precision for data cleaning and business-question answering with Power BI's strength in interactive, cross-filterable visual storytelling.

Together, they turn a static dataset into a living tool that stakeholders — job seekers, recruiters, and workforce planners alike — can explore on their own terms.

The project also demonstrates the importance of data validation and quality assurance by ensuring consistency between SQL query results and Power BI dashboard insights, resulting in reliable and accurate business reporting.

Overall, this project strengthened practical skills in SQL, Power BI, data modeling, dashboard design, and business insight generation, providing a strong foundation for solving real-world data analytics problems.

All underlying SQL queries, the cleaned dataset, and the full interactive Power BI dashboards are available alongside this report for further exploration and filtering by state, industry, company, or job title.